Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/01 Computer systems

Topic 1.6: Ethical, legal, cultural and environmental impacts

Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 80 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **73**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 73

1 The Data Protection Act 2018 controls how organisations use personal data.

(a) State what is meant by **personal data**. **[1]**

(b) A company stores the names and addresses of its customers. State **two** things the company must do to follow the Data Protection Act. **[2]**

(c) State **two** rights that the Data Protection Act gives to individuals. **[2]**

(d) A website asks users for their date of birth and phone number when they sign up. Explain one issue this could raise for the privacy of its users. **[2]**

2 Several laws apply to the use of computers.

(a) Draw **one** line from each law to the situation it protects. **[3]**

Law	**Protects against**
Computer Misuse Act 1990	Illegal copying of music, films and software
Copyright, Designs and Patents Act 1988	Personal data being misused by organisations
Data Protection Act 2018	Unauthorised access to computer systems

(b) Give **two** offences that are covered by the Computer Misuse Act. **[2]**

(c) A student copies a song from a friend's computer and sells copies online. State which law the student has broken. **[1]**

(d) A student guesses a teacher's password and reads the exam papers on the school network. State which law has been broken and explain why. **[2]**

(e) Give **two** ways copyright can be protected by the creator of software. **[2]**

3 Software can be released under different licences.

(a) Describe **one** difference between open source and proprietary software. **[2]**

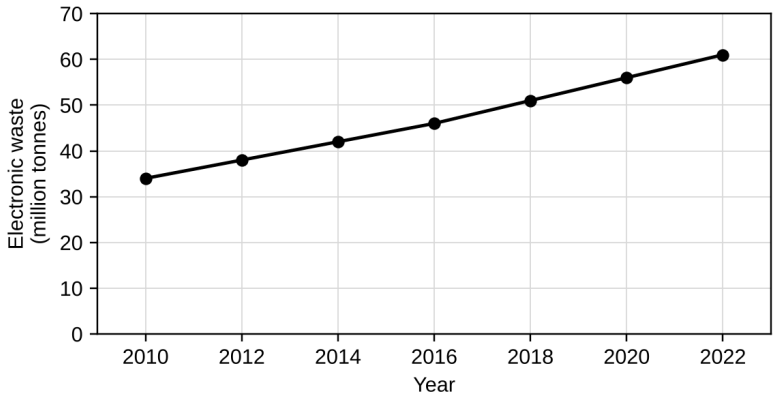
(b) Give **two** advantages of open source software. **[2]**

(c) Give **two** advantages of proprietary software. **[2]**

(d) A hospital must choose software to manage patient records. Recommend open source or proprietary software and justify your choice. **[3]**

4 Computers affect the environment.

(a) The graph shows the amount of electronic waste produced in the world between 2010 and 2022.



(b) Calculate the percentage increase in electronic waste from 2010 to 2022. **[3]**

Answer _____ %

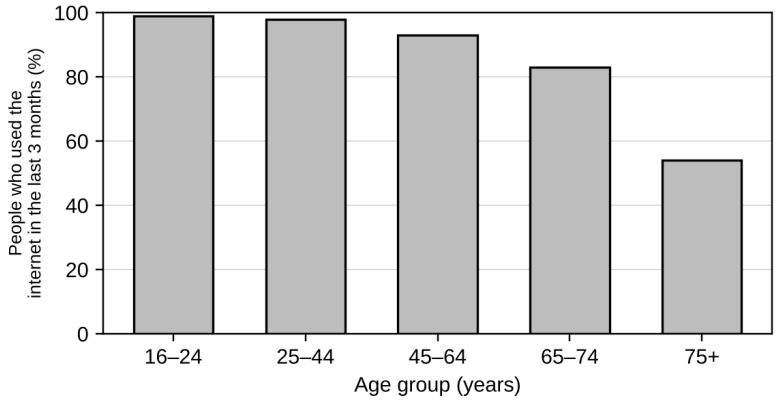
(c) Describe the trend shown in the graph. **[1]**

(d) Explain **two** reasons why e-waste is a problem for the environment. **[4]**

(e) Suggest **two** ways that e-waste can be reduced. **[2]**

(f) Data centres use large amounts of energy. Give **two** ways a company can reduce the energy used by its computers. **[2]**

5 The **digital divide** describes the gap between people who have access to technology and those who do not. The bar chart shows the percentage of people in a country in each age group who had used the internet recently.



(a) State the age group with the lowest internet use. **[1]**

(b) Calculate the difference between the highest and lowest percentages. **[1]**

Answer _____ %

(c) Give **two** reasons why older people may be less likely to use the internet. **[2]**

(d) Give **two** other factors that can lead to a digital divide. **[2]**

(e) Describe **one** way a government could reduce the digital divide. **[2]**

6 Technology raises many ethical and cultural issues.

(a) A company uses cameras with facial recognition in a shopping centre. Give **one** benefit and **one** concern. **[2]**

(b) A social media company collects data about its users' interests. Describe **two** ethical issues. **[2]**

(c) Driverless cars are controlled by software. Give **two** ethical or legal issues. **[2]**

(d) Explain how the use of computers and automation can affect employment. **[3]**

(e) Explain **one** way that online communication has changed the way people interact in society. **[2]**

7 Extended response questions.

(a) *A large company wants to introduce software that monitors every keystroke and website visited by its employees. Discuss the ethical, legal and privacy issues that this raises. **[8]**

(b) *Discuss the environmental impact of the increasing use of digital technology, and how it may be reduced. **[8]**

END OF QUESTIONS

MARK SCHEME – Topic 1.6: Ethical, legal, cultural and environmental impacts

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	information that identifies a living person (e.g. name, address, date of birth, medical record) ;	1 AO1 1.6.1	
1(b)	• keep data secure ; • keep data accurate and up to date ; • use data only for the stated purpose ; • only keep data for as long as needed ; • collect only the data necessary ; • process data lawfully, fairly and transparently ;	2 AO1 1.6.1	Any two.
1(c)	• to see the data held about them ; • to have inaccurate data corrected ; • to have data erased ; • to object to / restrict how their data is used ;	2 AO1 1.6.1	Any two.
1(d)	• data could be hacked / leaked / sold to third parties ; • used for identity theft / unwanted marketing ;	2 AO3 1.6.1	
2(a)	• Computer Misuse Act: protects against unauthorised access to computer material / hacking ; • Copyright, Designs and Patents Act: protects the creators' work from being copied without permission ; • Data Protection Act: protects individuals' personal data ;	3 AO1 1.6.1	One mark per correct match.
2(b)	• gaining unauthorised access to a computer / data ; • gaining unauthorised access with intent to commit a further crime (e.g. fraud) ; • unauthorised modification of data / spreading malware ;	2 AO1 1.6.1	Any two.
2(c)	Copyright, Designs and Patents Act (1988) ;	1 AO2 1.6.1	
2(d)	• Computer Misuse Act ; • unauthorised access to computer material ;	2 AO2 1.6.1	
2(e)	• use a licence agreement / EULA ; • product keys / activation codes ; • digital rights management (DRM) ; copyright notice ;	2 AO1 1.6.1	Any two.
3(a)	• open source: source code is available to view and modify ; • proprietary: source code is kept secret / cannot be changed by the user ;	2 AO1 1.6.1	Accept: open source is usually free; proprietary is usually paid for.
3(b)	• free / low cost ; • users can modify the code to fit their needs ; • bugs found and fixed quickly by a community ;	2 AO1 1.6.1	Any two.
3(c)	• supported / regularly updated by the company ; • more thoroughly tested / reliable ; • easier to use / documentation / training available ;	2 AO1 1.6.1	Any two.

Question	Answer	Marks	Guidance
3(d)	• either choice stated (must be justified) ; • e.g. proprietary: technical support / accountability / security updates guaranteed ; • or open source: lower cost / can customise to hospital's needs, but needs in-house skills ;	3 AO3 1.6.1	
4(b)	• 61 − 34 = 27 ; • 27 ÷ 34 × 100 ; • 79.4 (%) / 79 ;	3 AO2 1.6.1	
4(c)	e-waste has increased every year / steadily over time ;	1 AO2 1.6.1	
4(d)	• it contains toxic materials (lead, mercury) ; • which leak into the soil / water causing pollution ; • valuable / rare metals are wasted ; • so more mining / resource use is needed ;	4 AO2 1.6.1	Award 1 mark for each reason and 1 for each linked consequence, up to 2 reasons.
4(e)	• recycle old devices ; • repair / upgrade rather than replace ; • donate / sell second-hand devices ; • manufacturers design longer lasting products ;	2 AO3 1.6.1	Any two.
4(f)	• switch off / use sleep mode when not in use ; • use energy-efficient hardware ; • use renewable energy / better cooling ; virtualisation ;	2 AO3 1.6.1	Any two.
5(a)	75+ ;	1 AO2 1.6.1	
5(b)	99 − 54 = 45 (%) ;	1 AO2 1.6.1	
5(c)	• lack of skills / confidence ; • cost of devices / broadband ; • do not see the need / health or accessibility issues ; • limited access to training ;	2 AO3 1.6.1	Any two.
5(d)	• income / cost ; • geographic location (rural areas, poor broadband) ; • education ; • disability ;	2 AO1 1.6.1	Any two.
5(e)	• provide free / low-cost broadband or devices ; • e.g. in libraries / schools / community centres, or fund training ;	2 AO3 1.6.1	
6(a)	• benefit: helps to identify criminals / improves safety ; • concern: invades privacy / data could be misused / inaccurate identification ;	2 AO3 1.6.1	
6(b)	• users may not know / consent to how their data is used ; • data may be sold / used to manipulate opinions ; • data could be hacked ;	2 AO3 1.6.1	Any two.
6(c)	• who is responsible / liable in an accident (owner, manufacturer, programmer) ; • decisions the software makes in unavoidable collisions ; • loss of jobs (drivers) ; • safety / hacking risk ;	2 AO3 1.6.1	Any two.
6(d)	• some jobs are replaced by machines / automation ; • new jobs created (e.g. programming, maintenance) ; • workers may need retraining / skills ;	3 AO2 1.6.1	

Question	Answer	Marks	Guidance
6(e)	• can communicate instantly across the world / video calls ; • less face-to-face contact / increases cyberbullying / spread of misinformation ;	2 AO2 1.6.1	Accept any valid cultural effect with a clear explanation.
7(a)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • benefits: prevent misuse of work time / protect confidential data / detect cyber attacks / check compliance • privacy: employees' personal information may be recorded (passwords, messages) without consent; feeling of being watched may reduce trust and morale • legal: must follow the Data Protection Act 2018 / GDPR - data collected lawfully, for a stated purpose, kept secure, employees told • legal: unauthorised use or disclosure of data could breach the law; monitoring must be proportionate • ethical: balance between the company's right to protect itself and employees' right to privacy • security: stored monitoring data could be hacked; who can see it? • conclusion: monitoring can be justified if transparent, limited, and legal	8 AO1 / AO2 / AO3 1.6.1	

Question	Answer	Marks	Guidance
7(b)	**Level 3 (high), 6–8 marks**: Thorough knowledge and understanding of the topic with detailed, accurate and relevant points. A balanced, well-structured answer reaching a justified conclusion where asked. Technical terms used correctly throughout. **Level 2 (mid), 3–5 marks**: Reasonable knowledge and understanding. Some relevant points made, but the answer lacks detail, balance or a clear structure in places. **Level 1 (low), 1–2 marks**: Basic knowledge. A few simple, mostly isolated points with little or no development or linking. **Level 0, 0 marks**: No response or no response worthy of credit. **Indicative content** (credit any other valid points): • manufacturing devices uses rare metals / energy / water and causes pollution from mining • energy use of devices and data centres; carbon emissions if powered by fossil fuels • e-waste: toxic materials in landfill; short device lifespans; rapid upgrade cycle • positive impacts: video calls reduce travel; smart systems reduce energy use; paperless working • reduction: recycle, repair, upgrade, longer lasting products, renewable energy for data centres, energy efficient hardware, power management • government / company policies and laws (recycling regulations) and consumer behaviour • balanced judgement: benefits can outweigh costs only if managed properly	8 AO1 / AO2 / AO3 1.6.1	

Where marks are lost in Topic 1.6

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Name the law AND the reason:** 'Computer Misuse Act – unauthorised access'. A law without the link earns half marks.
- ☐ **Three laws:** Data Protection (personal data), Computer Misuse (hacking / malware), Copyright (illegal copying).
- ☐ **Open source ≠ no rules:** the code can be viewed and changed, but a licence still applies.
- ☐ **Impact questions:** give a positive *and* a negative, then relate them to the scenario in the question.
- ☐ **E-waste:** say what is harmful (toxic materials), where it goes (landfill / water) and what the effect is.
- ☐ **Digital divide:** give a cause (cost / skills / location) *and* an effect (cannot access services or jobs).
- ☐ **Data interpretation:** percentage change = change ÷ original × 100. Show the working.
- ☐ **Discuss:** balanced conclusion at the end ('overall...') is needed for top marks.